

Conceptualizing semantic transparency: A systematic analysis of semantic transparency measures in English compound words[☆]

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ABSTRACT

Semantic transparency refers to the extent to which the meaning of a multimorphemic word can be determined from the meaning of its constituents. In terms of compounds, semantic transparency has been operationally defined in multiple ways, and research has sometimes produced conflicting results concerning its effect on processing. This study explored a potential source of these experimental conflicts by examining the methodology and measures used to operationalize semantic transparency. First, we used factor analysis to investigate whether common measures of semantic transparency truly inform the same underlying construct and found that there were at least four factors represented by eleven common semantic transparency variables. After extracting predicted values from these factors, we found that different aspects of semantic transparency are not equally predictive of behavioural data from lexical decision and naming tasks. Moreover, the various aspects represented by the different measures appear to interact and influence each other when predicting this behavioural data. We conclude that different measures of semantic transparency reflect different constructs, and that this should be considered when investigating the effect of semantic transparency.

Semantic transparency is a theoretical construct referring to the extent to which the constituents of multi-morphemic words (e.g., *fool* + *ish* or *re* + *hearse* in the case of derived words, or *snow* + *ball* or *shin* + *dig* in the case of compound words) contribute to the meaning of the whole word. Semantic transparency ranges continuously from fully transparent to fully opaque. For example, the compound *snowball* is considered to be transparent because it derives its meaning from both *snow* and *ball*, whereas *shindig* is considered to be opaque because its meaning is unrelated to the meaning of *shin* or *dig*. Other compounds, such as *strawberry*, are semantically transparent with respect to one constituent (e.g., *berry*) but opaque with respect to the other (e.g., *straw*). Semantic transparency has been defined and measured several different ways in the context of compound words (see Gagné et al., 2019, and Gagné et al., 2016, for discussion), each of which might tap into different aspects of semantic transparency. Our goal in this paper is to understand the relationships among these different measures, and how these operational definitions relate to the broader theoretical construct. We begin by discussing the various measures that are common in the literature, and we then use factor analysis to identify the potential underlying constructs targeted by these measures. After obtaining this set of factors, we will

extract values of these factors for a large number of compound words and conduct regression analyses to evaluate how the underlying constructs may be predictive of traditional behavioural measures of human performance, such as lexical decision response time data.

Semantic transparency has figured prominently in research on compound word representation and processing. Compound words are the linguistic representation of a combined concept and serve as the midpoint between words and phrases due to their having two constituents (e.g., *snow* + *ball*) that must be integrated into a linguistic structure (Libben, 2006; Gagné and Spalding, 2013). Previous research on compounds shows mixed results regarding the extent to which semantic transparency influences processing. Some studies have shown that semantic transparency influences ease of processing across multiple paradigms, including eye-tracking or EEG (Brunsighan and Folk, 2012; Juhasz, 2007; Schmidtke et al., 2018; Davis et al., 2019) and semantic priming and/or lexical decision tasks (Sandra, 1990; Zwitserlood, 1994; El-Bialy et al., 2013; Günther and Marelli, 2019; Kim et al., 2018; Marelli and Luzzatti, 2012; Libben et al., 2003). Conversely, other studies have shown that semantic transparency has no effect on ease of processing (i.e., the processing of semantically transparent and opaque compounds

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did not differ) within these same paradigms and others (Juhász, 2018; Pollatsek and Hyönä, 2005; Smolka and Libben, 2017; Gummior et al., 2006; Fiorentino and Fund-Reznicek, 2009; Dohmes et al., 2004). Others have shown that the effects of semantic transparency become larger when the compositionality of the compound is emphasized (Ji et al., 2011; Frisson et al., 2008; Marelli et al., 2015; Kuperman and Bertram, 2013; Inhoff et al., 2000).

Compositionality refers to the extent to which a word's meaning can be determined from its constituents. For example, the meaning of *raining* can be derived from the meaning of the stem *rain* and the suffix *-ing*. In the case of compound words, there is a wide range of compositionality, and it is rare that the meaning of a compound is fully compositional; although the meaning of *snowman* is clearly related to the meaning of its morphological constituents *snow* and *man*, there are many aspects of the meaning (such as snowmen tending to have eyes made of rocks or coal) that cannot be derived from the parts. A key observation concerning compositionality is that the constituents play different roles, in that the final constituent in English compounds denotes the syntactic head of the compound and the first constituent functions as the modifier (Libben et al., 2003; Gagné and Shoben, 1997; Schreuder and Baayen, 1995). Whether the syntactic head corresponds to the semantic category of the compound depends on whether the second constituent is semantically transparent. To illustrate, consider the compounds *logbook*, *notebook*, *teacup*, and *buttercup*. *Log* in *logbook* is more opaque than *note* in *notebook*, but both *logbook* and *notebook* are different members of the category 'book'. Conversely, *cup* in *buttercup* is more opaque than *cup* in *teacup*, and this changes the semantic category of *buttercup* (i.e., *buttercup* is a member of the semantic category 'flower' not 'cup'). Thus, the transparency of the second constituent (in English) influences the semantic category, while the transparency of the first constituent does not. We will return to the relevance of this distinction in the General Discussion.

Despite its prominent role in understanding compound processing, and potentially related to the conflicting results in previous studies, semantic transparency is often defined differently across studies. Particularly, semantic transparency has been defined as an association of compound and constituent meaning (e.g., Schmidtke et al., 2018b; Kim et al., 2018; Wang et al., 2014), the retention of constituent meaning within the compound (e.g., Gagné et al., 2019; Marelli and Luzzatti, 2012), or the predictability of compound meaning from its constituents (e.g., Gagné et al., 2019; Marelli and Luzzatti, 2012; Juhász et al., 2015). Unsurprisingly, given these multiple definitions, researchers have used several different ways of operationalizing semantic transparency. Furthermore, common operational measures can also be categorized as either computationally derived from distributed semantic models, or obtained experimentally from human ratings of transparency.

Distributed semantic models measure word-level semantic association and relation as a function of co-occurrence and use distances between high-dimensional word vectors calculated from corpus data (e.g. Marelli et al., 2015; Schmidtke et al., 2018; Gagné et al., 2016). These corpus-based association values are indicative of semantic similarity (Landauer and Dumais, 1997; Mandera et al., 2017; Ji et al., 2011; Marelli and Luzzatti, 2012). Latent Semantic Analysis (LSA, Landauer and Dumais, 1997) and SNAUT (Mandera et al., 2017) are two distributed semantic models commonly used to describe semantic transparency, and there are theoretical, and consequently computational, differences between the two. LSA is considered a *count* model, because it initially counts the number of occurrences of a word in a given text to represent the contexts in which the word typically occurs (Mandera et al., 2017). Associations between words are then calculated based on the similarity of their respective contexts (Landauer and Dumais, 1997; Mandera et al., 2017). Conversely, SNAUT is considered a prediction-based model, because the system is based on a semantic network and implicitly learns to predict an 'event', such as the target word, based on the presence of other words (i.e., the context) in the given text (Mandera et al., 2017; Kim et al., 2018). Comparable to count models, the association between two words in prediction models is derived from the similarity between the

contexts in which these words are predicted to occur.

A primary distinction between the two types of models is the inclusion of negative sampling, which SNAUT possesses but LSA does not (Landauer and Dumais, 1997; Mandera et al., 2017). Negative sampling accounts for negative associations between words and creates a more complete picture of the semantic context by accounting for words that do not occur with the target word (Johns et al., 2019). While inclusion of negative sampling improves the representation of word similarity compared to models in which negative sampling is absent (Johns et al., 2019; Baroni et al., 2014), researchers often still use both types of models in psycholinguistics (e.g., Kim et al., 2018; Schmidtke et al., 2018; Wang et al., 2014; Mandera et al., 2017). Thus, both SNAUT and LSA are useful for the current project in determining the nature of semantic transparency as it is currently conceived in the literature.

Distributed semantic models, such as LSA and SNAUT, calculate the association between word meanings based on experiential learning from text and in a manner comparable to human judgements (Landauer and Dumais, 1997). These measures have been used in the literature as a proxy for semantic transparency because transparent constituents tend to be more associated to the compound than are opaque constituents both by the model and by a human participant (e.g., *blueberry* is more associated to *berry* or *blue* than *shindig* is to *dig* or *shin*). Consequently, models such as LSA and SNAUT are useful in providing measures that appear to be good approximations of semantic transparency. When used to operationalize semantic transparency, these values are often found between the first constituent and the compound, the second constituent and the compound, and the two constituents.

Human ratings are based on human perception of the interaction between constituent and compound meaning, and obtained experimentally prior to a research study. Human ratings are quite sensitive to differences in semantic transparency (Kim et al., 2018). How these ratings are obtained can differ, and may depend on the definition of semantic transparency to which the researcher subscribes (Gagné et al., 2016). Additionally, researchers have used different scale types and scale ranges to ask participants to describe the level of transparency. A singular rating scale, also known as a Likert item, is most commonly used, but researchers use differing ranges across studies (e.g., Libben et al., 2003 vs. Juhász et al., 2015) or even continuous scales such as percentage (e.g., Gagné et al., 2019). The Large Database of English Compounds (LADEC; Gagné et al., 2019) contains semantic transparency variables representing all three theoretical perspectives of semantic transparency mentioned previously. Specifically, in this study we look at human rating variables of constituent meaning retention within the compound, compound predictability from the constituents, and relatedness between each constituent and the compound. Clearly, there are various measures used by different researchers to represent semantic transparency, which may be related to the conflicting results regarding this construct: One possible explanation for at least some of these inconsistent findings is that different measures were used to operationalize semantic transparency across the different studies.

As an example of how different operational measures may inform different aspects of semantic transparency, and thus cause conflicting experimental results, we will look at the compounds *snowball* and *quarterback*. Human ratings generally indicate that the meaning of *snowball* is highly predictable from the meanings of *snow* and *ball*, and these constituents have high meaning retention in the compound *snowball*. However, *snowball* is closely associated with *snow*, but not necessarily with *ball*, as shown by the corpus-based computational measures of association from LSA and SNAUT. The relationship between measures is even more complex for items such as *quarterback*, which are in the middle of the opaque-transparent continuum. Human ratings from LADEC indicate that the meaning of the compound *quarterback* is moderately predictable from the meaning of its constituents, but the ratings for meaning retention for both constituents are lower than its predictability, and measures of association between *quarterback*, *quarter*, and *back* are even lower. Thus, depending on which perspective of semantic transparency is

prioritized, *quarterback* may be considered more opaque or more transparent. This discrepancy may indicate that different measures of semantic transparency are mapping onto different theoretical constructs (or different aspects of the theoretical construct) of semantic transparency, which may not be equivalent. Therefore, examining and identifying these potential constructs is crucial for enhancing our understanding of how semantic transparency influences word processing.

1. The current study

Our goal is to examine the methodology that has been used thus far to study semantic transparency. More specifically, we will determine whether the various measures of semantic transparency used commonly in psycholinguistics map onto the same theoretical construct or whether they reflect different aspects of semantic transparency. We will explore the underlying structure of semantic transparency through the measures used as its proxy, and attempt to investigate the complex facets of this construct that may be represented by different measures. We will conduct factor analysis on operational measures representative of those commonly used in the literature, to summarize the structure of their interrelationships. Then, we will apply this structure to the conceptualization of semantic transparency and its influence on human processing using behavioural data and regression.

2. Identifying the factor structure of semantic transparency

Like any theoretical construct, semantic transparency cannot be measured directly; researchers instead rely on various operational measures. To our knowledge, semantic transparency has not been investigated by factor analysis, a method common in many areas of psychology, especially social psychology and educational psychology. Factor analysis is useful for moving beyond operational variables or concrete values and providing a picture of the abstract concept being investigated. Specifically, it can help parse the differences between different constructs or between different aspects of the same construct. Therefore, we will use exploratory factor analysis to identify the structure of relations among the various measures of semantic transparency and to reduce the larger set of variables into a set of factors (Gorsuch, 1983; Kline, 1994). Each factor provides "... a condensed statement of the relationships between a set of variables" (Kline, 1994, p. 5). Thus, this method is well suited for providing insight into how the measures are related to each other, and, consequently, is a useful method for understanding the nature of

semantic transparency as a construct. We use exploratory methods, rather than confirmatory factor analysis (e.g., Suhr, 2006), because the underlying structure of semantic transparency is not previously known and cannot be reliably hypothesized (Gorsuch, 1983).

2.1. Methods

2.1.1. Materials

For our investigation, we selected eleven measures of semantic transparency which are representative of the different perspectives of semantic transparency. Five were different human ratings of transparency related to the compound and its constituents (see Gagné et al., 2019 and Kim et al., 2018 for details); one represents compound predictability (LADEC variable: ratingcmp), two represent meaning retention of the first and second constituents (LADEC variables: ratingC1, ratingC2), and two represent meaning relatedness of each constituent with the compound (LADEC variables: st_c1_mean, st_c2_mean). The other six represented meaning associations and were obtained from the distributed semantic models LSA and SNAUT. LSA and SNAUT values were found between the first constituent and the compound (LADEC variables: LSAc1stim, c1stim_snaucos), the second constituent and the compound (LADEC variables: LSAc2stim, c2stim_snaucos), and the constituents themselves (LADEC variables: LSAc1c2, c1c2_snaucos). Only items with values for all eleven variables were used, resulting in a sample size of 1555 compounds. The descriptions of each variable and its descriptive statistics are presented in Table 1. Fig. 1 shows the histograms of the meaning retention and predictability variables for these compounds and their constituents (Gagné et al., 2019).

2.1.2. Procedure

We used exploratory factor analysis to investigate the latent structure of semantic transparency measures. Data were analyzed in Stata 15 (StataCorp, 2017b). Factors were extracted using principal factor analysis under the common factor model, because this model crucially differentiates between common, unique, and error variance, and does not require high communalities (Kline, 1994; Ford et al., 1986; Fabrigar et al., 1999). Factor loadings were considered significant if their absolute values were above 0.32 (Yong and Pearce, 2013; Osborne et al., 2008). Factors were retained based on two criteria: Cattell's scree test (Cattell, 1966, 1978) and Horn's parallel analysis (Horn, 1965; Dinno, 2009). Parallel analysis was used to designate the upper bound for the possible number of factors to retain (Ford et al., 1986), and the scree plots were

Table 1

Descriptive statistics and descriptions of the eleven semantic transparency variables for the items used in the current project. Variables were obtained from the Large Database of English Compounds (LADEC; Gagné et al., 2019). Original sources of the respective measures are listed under Source.

	LADEC Variable Name	Description	Unit/Scale/Range	Observations	Mean	Std. Dev.	Min	Max	Source
Human Ratings	ratingcmp	Predictability of compound from constituents (C1 & C2)	%	1555	65.77	16.80	22.19	96.87	Gagné et al. (2019)
	ratingC1	Meaning retention of first constituent (C1) in compound	%	1555	68.62	19.74	6.50	96.62	
	ratingC2	Meaning retention of second constituent (C2) in compound	%	1555	72.16	17.52	1.13	96.79	
	st_c1_mean	Relatedness between C1 and compound	1 to 7	1555	4.17	0.81	1.75	6.39	Kim et al. (2018)
	st_c2_mean	Relatedness between C2 and compound	1 to 7	1555	3.89	0.80	1.78	6.53	
Distributed Semantic Models	LSAc1c2	LSA between C1 and C2	-1 - 1	1555	0.21	0.15	-0.04	0.92	Landauer & Dumais (1997)
	LSAc1stim	LSA between C1 and compound	-1 - 1	1555	0.19	0.19	-0.10	0.90	
	LSAc2stim	LSA between C2 and compound	-1 - 1	1555	0.17	0.16	-0.08	0.92	Mandera et al. (2017)
	c1c2_snaucos	SNAUT cosine distance between C1 and C2	0 - 2	1555	0.74	0.12	0.21	1.06	
	c1stim_snaucos	SNAUT cosine distance between C1 and compound	0 - 2	1555	0.73	0.13	0.33	1.08	
	c2stim_snaucos	SNAUT cosine distance between C2 and compound	0 - 2	1555	0.73	0.13	0.24	1.05	

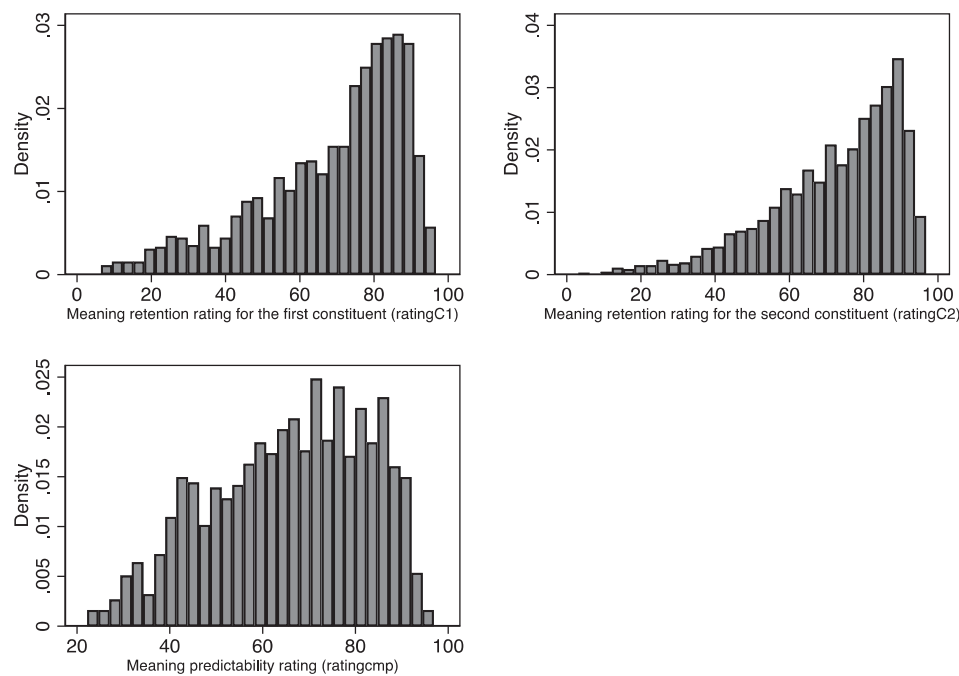


Fig. 1. Histogram of values for the LADEC human rating measures of semantic transparency from Gagné et al. (2019) for the items used in the current project.

the primary diagnostic criterion (Cattell and Vogelmann, 1977). Models including a factor that had one or fewer significant loadings after rotation were excluded (Osborne et al., 2008). Oblique rotation was used to facilitate interpretation of the factor models. Presumably, the factors extracted from our data should designate different facets of a single construct; therefore, these factors should be correlated and oblique rotation is preferred over orthogonal (Kline, 1994). The specific method of rotation was promax, and power was 3 (Hendrickson and White, 1964).

Regarding our assumptions, the Kaiser-Meyer-Olkin (KMO) measure of sampling adequacy for all variables was 0.723, which is sufficient to indicate our sample is appropriate for exploratory factor analysis (Kaiser, 1970). That is, there is sufficient shared variance within our data, meaning that they are indeed psychometrically related, but not to the degree of multicollinearity (Beavers et al., 2013; Kaiser, 1970). Table 2 shows the correlation matrix between all eleven variables. The determinant of the correlation matrix including all variables was 0.003. Only six correlations had a magnitude between 0.60 and 0.75. The other 49 correlations were below this threshold. As the mean correlation magnitude was 0.30 and the determinant was non-zero, this correlation matrix

indicates an absence of multicollinearity (Yong and Pearce, 2013). The Bartlett's test of sphericity was significant ($p < 0.0001$), indicating that our data had sufficient intercorrelations, or enough of a patterned relationship, to suggest that factor analysis will produce viable factors (Bartlett, 1950; Yong and Pearce, 2013). Addressing sample size, most scholars suggest an item-variable ratio of 5:1 or 10:1 (Gorsuch, 1983; Osborne et al., 2008). The current study has an approximate item-variable ratio of 141:1 and a sample size of 1555, and so should be sufficient to represent the population of English noun-noun compounds.

2.2. Analyses and discussion

Our analysis includes three distinct Factor Models, all with the same items, and method of extraction, rotation, and retention. Model 1 includes eleven semantic transparency variables shown in Table 1. In Model 2 we remove two variables because their loadings suggest they uniquely inform a separate factor from the other nine variables. Model 3 is a two-factor model, meant to examine whether our data is primarily distinguished by the related constituent (i.e., the first constituent or the second constituent), or by type of measure (i.e., corpus-based

Table 2

Correlation matrix for the eleven semantic transparency variables for the items used in the current project. Variables include: human ratings related to the first constituent (ratingC1, st_c1_mean), the second constituent (ratingC2, st_c2_mean), the compound itself (ratingcmp); computational associations between the first constituent and the compound (LSAc1stim, c1stim_snautCos), the second constituent and the compound (LSAc2stim, c2stim_snautCos), and the two constituents (LSAc1c2, c1c2_snautCos).

	ratingcmp	ratingC1	ratingC2	st_c1_mean	st_c2_mean	LSAc1c2	LSAc1stim	LSAc2stim	c1c2_snautCos	c1stim_snautCos
ratingcmp	1.00									
ratingC1	0.75	1.00								
ratingC2	0.66	0.24	1.00							
st_c1_mean	0.62	0.75	0.24	1.00						
st_c2_mean	0.49	0.17	0.72	0.26	1.00					
LSAc1c2	0.21	0.12	0.10	0.11	0.13	1.00				
LSAc1stim	0.35	0.40	0.16	0.44	0.16	0.17	1.00			
LSAc2stim	0.24	0.08	0.33	0.08	0.40	0.26	0.27	1.00		
c1c2_snautCos	-0.27	-0.15	-0.18	-0.21	-0.24	-0.65	-0.16	-0.20	1.00	
c1stim_snautCos	-0.39	-0.44	-0.18	-0.52	-0.22	-0.16	-0.56	-0.12	0.30	1.00
c2stim_snautCos	-0.24	-0.01	-0.42	-0.07	-0.53	-0.16	-0.15	-0.50	0.29	0.31

computational models or human rating). We used these three models because each subsequent analysis allows us to explore the variable relationships within our data and their consistency across analyses. The first model establishes the pattern of factor loadings, the second model further specifies and validates this model, and the third model explores the overarching pattern followed by the data.

2.2.1. Model 1: including all eleven semantic transparency variables

The results from this analysis indicated that the eleven semantic transparency variables inform at least four separate factors or aspects of semantic transparency. These factors were retained based on the results of the scree plot shown in Fig. 2. Horn's parallel analysis indicated the retention of six factors as an upper bound, but the fifth and sixth factor models included factors with single or zero rotated loadings and were automatically excluded. Table 3 shows the loadings corresponding to each rotated factor.

The finding that the variables load onto four distinct factors support our hypothesis that the various measures of semantic transparency reflect different aspects of this construct.

In particular, the first three factors reflect differences between constituents and between the human ratings and corpus-based association measures, and Factor 4 shows that LSA and SNAUT values of association between the constituents (excluding the compound) load separately from the other variables.

The factor loadings on the first two factors are distinct; Factor 1 is comprised only of measures related to the first constituent, while Factor 2 is comprised only of only measures related to the second constituent. This suggests that each constituent separately contributes to the overall construct of semantic transparency. The predictability rating of the

compound from its constituents is cross-loaded between these two factors, which corroborates the distinction as this measure would include contributions from both constituents. Factor 3 appears to be a computational factor (i.e., it reflects only the corpus-based association measures of semantic transparency), as it does not contain any of the human rating variables. We propose that this provides further evidence that human ratings and corpus-based association measures may not have identical, or interchangeable, contributions to the semantic transparency of a compound (Gagné et al., 2019).

The factor loadings on the first three factors suggest that the constituents contribute to semantic transparency differently not only at a base level, but how the constituents contribute also differs. Factors 1 and 3 show that for the first constituent, the human ratings and corpus-based computational measures for semantic transparency load separately, while Factor 2 shows that these types of measures load together for the second constituent. This suggests that there is a difference between how the first and second constituents influence the semantic transparency of the compound, and that each constituent uniquely informs separate aspects of semantic transparency.

Factor 4 contains only the LSA and SNAUT values for the association between the constituents themselves, excluding the compound (LSAc1c2; c1c2_snaucos). All other variable loadings on Factor 4 were less than $|0.10|$, and the loadings of these two variables on the other 3 factors were less than $|0.08|$. This suggests that the association between the first and second constituent is distinct from measures of semantic transparency related to the compound. If this is the case, it is possible that these two variables are uniquely targeting a construct that the other variables are not. To examine this possibility, we performed an analysis (Model 2 reported in the subsequent section) without these variables to determine whether removing these LSA and SNAUT variables influences the factor structure.

2.2.2. Model 2: excluding LSA and SNAUT values between the first and second constituent

In this analysis, we excluded the LSA and SNAUT measures of association between the first and second constituent to investigate the stability and validity of Factors 1–3 in Model 1. If the LSA and SNAUT values were uniquely targeting a separate construct, their removal should not greatly affect the loadings of the other variables.

Model 2 included the nine variables which informed the first three factors of Model 1. The value from the Kaiser-Meyer-Olkin (KMO) test for this set of variables was 0.73, which indicates that our data was adequate for Factor Analysis. The Bartlett's test for sphericity was significant ($p < 0.0001$), and the determinant of the correlation matrix was 0.007. Table 4 shows the rotated factor loadings. Three factors were retained based on the scree plot (Fig. 3).

Clearly, the factor structure is consistent across Models 1 and 2, even when the corpus-based association measures between the first constituent and second constituent are excluded. Not only does this suggest our

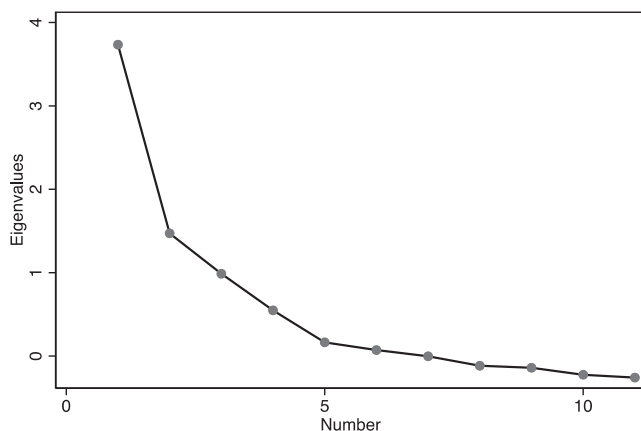


Fig. 2. Scree plot of the eigenvalues indicating the number of factors to retain for Factor Model 1, which includes eleven semantic transparency variables.

Table 3

Rotated Factor loadings from Factor Model 1, retaining four factors and containing eleven semantic transparency variables. Variables include: human ratings related to the first constituent (ratingC1, st_c1_mean), the second constituent (ratingC2, st_c2_mean), the compound itself (ratingcmp); computational associations between the first constituent and the compound (LSAc1stim, c1stim_snaucos), the second constituent and the compound (LSAc2stim, c2stim_snaucos), and the two constituents (LSAc1c2, c1c2_snaucos). Significant loadings (greater than $|0.32|$) are marked with *.

Variable	Factor1	Factor2	Factor3	Factor 4	Uniqueness
ratingcmp	0.7418*	0.4654*	−0.0889	0.0818	0.1395
ratingC1	0.8566*	−0.0148	0.1034	0.0049	0.2093
ratingC2	0.2292	0.8668*	−0.1516	−0.0642	0.2581
LSAc1c2	0.0195	−0.0764	−0.0375	0.7764*	0.4509
LSAc1stim	0.2590	−0.0765	0.5962*	−0.0406	0.5447
LSAc2stim	−0.1647	0.4067*	0.2789	0.0959	0.6376
c1c2_snaucos	−0.0483	−0.0011	−0.019	−0.7465*	0.4169
c1stim_snaucos	−0.2910	0.0699	−0.6288*	−0.0052	0.4485
c2stim_snaucos	0.2353	−0.5327*	−0.3373*	−0.0434	0.4704
st_c1_mean	0.7044*	−0.0193	0.2780	−0.0164	0.3313
st_c2_mean	0.0644	0.7869*	0.0446	−0.0354	0.3516

Table 4

Rotated factor loadings from Factor Model 2, retaining three factors and containing nine semantic transparency variables. Variables include: human ratings related to the first constituent (ratingC1, st_c1_mean), the second constituent (ratingC2, st_c2_mean), and the compound itself (ratingcmp); computational associations between the first constituent and the compound (LSAc1stim, c1stim_snaucos), and the second constituent and the compound (LSAc2stim, c2stim_snaucos). Significant loadings (greater than |0.32|) are marked with *.

Variable	Factor1	Factor2	Factor3	Uniqueness
ratingcmp	0.7430*	0.4849*	-0.0628	0.1482
ratingC1	0.8613*	-0.0096	0.0887	0.2100
ratingC2	0.2270	0.8472*	-0.1551	0.2639
LSAc1stim	0.2703	-0.0952	0.5771*	0.5440
LSAc2stim	-0.1564	0.4127*	0.3302*	0.6460
c1stim_snaucos	-0.3072	0.0774	-0.6109*	0.4615
c2stim_snaucos	0.2273	-0.5242*	-0.3734*	0.4761
st_c1_mean	0.7141*	-0.0237	0.2525	0.3329
st_c2_mean	0.0658	0.7708*	0.0514	0.3528

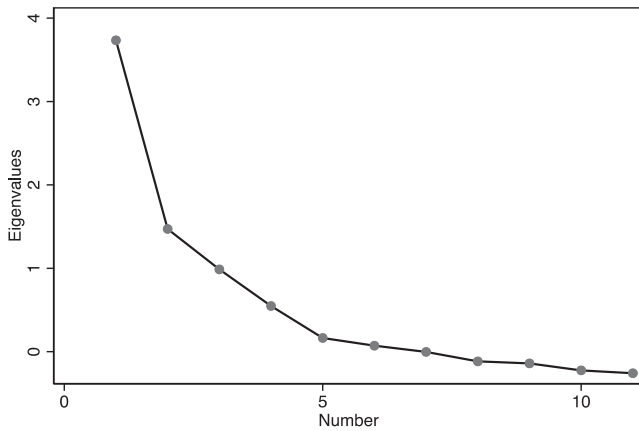


Fig. 3. Scree plot of the eigenvalues indicating the number of factors to retain for Factor Model 2, which includes nine semantic transparency variables.

data is stable across analyses, but it supports the results of Model 1. The first three factors are virtually identical, so it is quite possible that the corpus-based computational measures for the relationship between the constituents do not target the same aspect of semantic transparency. Human ratings and corpus-based computational measures related to the first constituent load separately, while human ratings and corpus-based computational measures related to the second constituent load together. Additionally, all four corpus-based variables used in this model now load significantly onto Factor 3, which bolsters its interpretation as a computational association factor. This means that the corpus-based variables related to the second constituent and the compound are now both cross-loaded, and the primary loading is still on Factor 2. Moreover, the highest variable loadings for Factor 3 are still the LSA and SNAUT values between the first constituent and the compound, which means that the first constituent is disproportionately represented in this Factor compared to the second constituent. These constituent differences will be discussed further in the General Discussion.

Although Models 1 and 2 show finer distinctions between the types of measures used to describe semantic transparency, it is unclear what the fundamental distinction between the variables might be. If we force a two-factor model, creating two groups of variables, how will the variables be organized? Will they separate based on which constituent they are related to? Or, will they separate based on the type of measure, human rating vs computational? Model 3 explores these questions.

2.2.3. Model 3: forcing a two-factor model

In this factor model, we included only the nine variables present in Model 2. Table 5 shows the rotated factor loadings. Before discussing further, note that this model shows an *incorrect* factor solution as there

are too few retained factors (see Fig. 3). Thus, it provides no basis for definitive hypotheses regarding the optimum underlying structure for these variables; however, this model is still useful for addressing our questions concerning the initial distinction between the variables. The factor loadings show the variables have split based on the constituent to which they are related. This suggests that the position of the constituent, rather than the type of measure, may be the rudimentary distinction which differentiates the contribution of semantic transparency variables.

2.2.4. Interim summary

In the preceding analyses, we identified two viable factor models to describe the relationship between variables representing Semantic Transparency. In Model 1 we found four distinct factors representing the relationships between eleven variables, and each factor appears to represent a single construct. Factor 1 represents the human ratings of the first constituent contributions to transparency, Factor 2 represents both the computational association and human ratings of second constituent contributions to transparency, Factor 3 represents the computational association between the constituents and the compound, and Factor 4 represents the computational association between the constituents only.

In Model 2, we removed the variables that loaded onto Factor 4 and ran an identical factor analysis procedure because the Factor 4 variables were targeting a construct not represented by the other factors. In this second model, the composition of the first three factors are identical to that of Model 1, which indicates that the relations between these nine variables are relatively independent of the variables that load onto the fourth Factor. Because of this separation, we will use the factors obtained in Model 2 for the analyses in the following sections.

Model 3 was forced to extract two factors from the nine variables in Model 2 and thus, allowed us to evaluate the fundamental distinction between these variable relationships, which we found was the position of the constituent (i.e., first vs. second) rather than the type of measure (human judgment vs. corpus-based).

3. Implementing the factor structure

Having obtained a factor model with viable factors, we were able to address the question of whether these factors predict human processing behaviour. In this section, we evaluate whether the factors identified in Model 2 (the most valid and viable model) predict response time data from lexical decision and naming tasks. Model 2 was used for these analyses because Factor 4 in Model 1 reflected a very different construct (i.e., it reflects only the corpus-based variables reflecting the relationship between the two constituents and not any relationship to the compound), which may have complicated further analyses (see 2.2.4 Interim Summary). Because each factor is a composite of the interrelationships between variables, they may form a more expansive, or comprehensive,

Table 5

Rotated Factor loadings from Factor Model 3, retaining two factors and containing nine semantic transparency variables. Variables include: human ratings related to the first constituent (ratingC1, st_c1_mean), the second constituent (ratingC2, st_c2_mean), and the compound itself (ratingcmp); computational associations between the first constituent and the compound (LSAc1stim, c1stim_snaucos), and the second constituent and the compound (LSAc2stim, c2stim_snaucos). Significant loadings (greater than |0.32|) are marked with *.

Variable	Factor1	Factor2	Uniqueness
ratingcmp	0.6731*	0.3582*	0.2466
ratingC1	0.9070*	-0.1121	0.2374
ratingC2	0.0776	0.7759*	0.3489
LSAc1stim	0.5348*	0.0639	0.6856
LSAc2stim	-0.0529	0.5621*	0.7025
c1stim_snaucos	-0.5850*	-0.0883	0.6132
c2stim_snaucos	0.1162	-0.7022*	0.5516
st_c1_mean	0.832*	-0.0460	0.3329
st_c2_mean	0.0135	0.7954*	0.3594

construct of semantic transparency than could be captured by a single variable. Thus, the factors may be useful to evaluate the targeted influence of their reflected construct, such as the semantic transparency of the second constituent, on the processing of compound words.

3.1. Methods

3.1.1. Materials

For each of the 1555 compounds in our sample, we created a new set of variables which were the predicted estimates for each factor from Model 2. These values were obtained using the regression (also called Thomson) scoring method, which is more accurate than the Barlett method, and are the weighted sum of the variables that load onto that factor (StataCorp, 2017a). Fig. 4 shows the histograms for the predicted values of each extracted factor for our set of items. For all three extracted factors, more negative values are associated with lower scores of transparencies (i.e., less transparent), while more positive values are associated with higher scores of transparencies (i.e., more transparent).

As our dependent variables, we used three lexical decision or naming variables measuring response time present in LADEC for these analyses (Gagné et al., 2019). These variables were obtained from the English Lexicon Project (ELP; Balota et al., 2007) and the British Lexicon Project (BLP; Keuleers et al., 2012). Specifically, we used lexical decision times from the BLP and ELP (LADEC variables: BLPr_t and elp_ld_rt), and naming latencies from the ELP (LADEC variable: elp_naming_mean_rt). Only items with ELP and BLP response times were used, resulting in a sample size of 1054 compounds for the BLP regressions and 1555 compounds for the ELP regressions. Table 6 shows the descriptive statistics for this subset of compounds in all three response time variables from the ELP

Table 6

Descriptive statistics of the lexical decision times from the British Lexicon Project (BLPr_t) and English Lexicon Project (elp_ld_rt), and naming latencies from the English Lexicon Project (elp_naming_mean_rt) reported for the items used in the current project.

Variable	Obs	Mean	Std. Dev.	Min	Max
BLPr_t	1054	666	76	466	1071
elp_ld_rt	1555	773	109	534	1447
elp_naming_mean_rt	1555	696	78	554	1064

and BLP.

3.1.2. Procedure

In these analyses, we had two sets of predictors: the nine semantic transparency variables present in Model 2 (see Table 1) and the three extracted factors from Factor Model 2 (see Table 4). We conducted a series of regression analyses in which these sets of variables were entered as predictor variables. Separate analyses were conducted for each of the three dependent variables (i.e., lexical decision latencies from ELP, lexical decision latencies from BLP, and naming latencies from ELP). Data were analyzed in Stata 15 (StataCorp, 2017b).

3.2. Analyses and discussion

3.2.1. Analysis 1: semantic transparency variables as predictors

First, we analyzed whether all nine variables used to create Factor Model 2 are predictive of behavioural response time data. Although subsets of the nine semantic transparency variables have been shown to be predictive of this type of behavioural data (e.g., Gagné et al.,

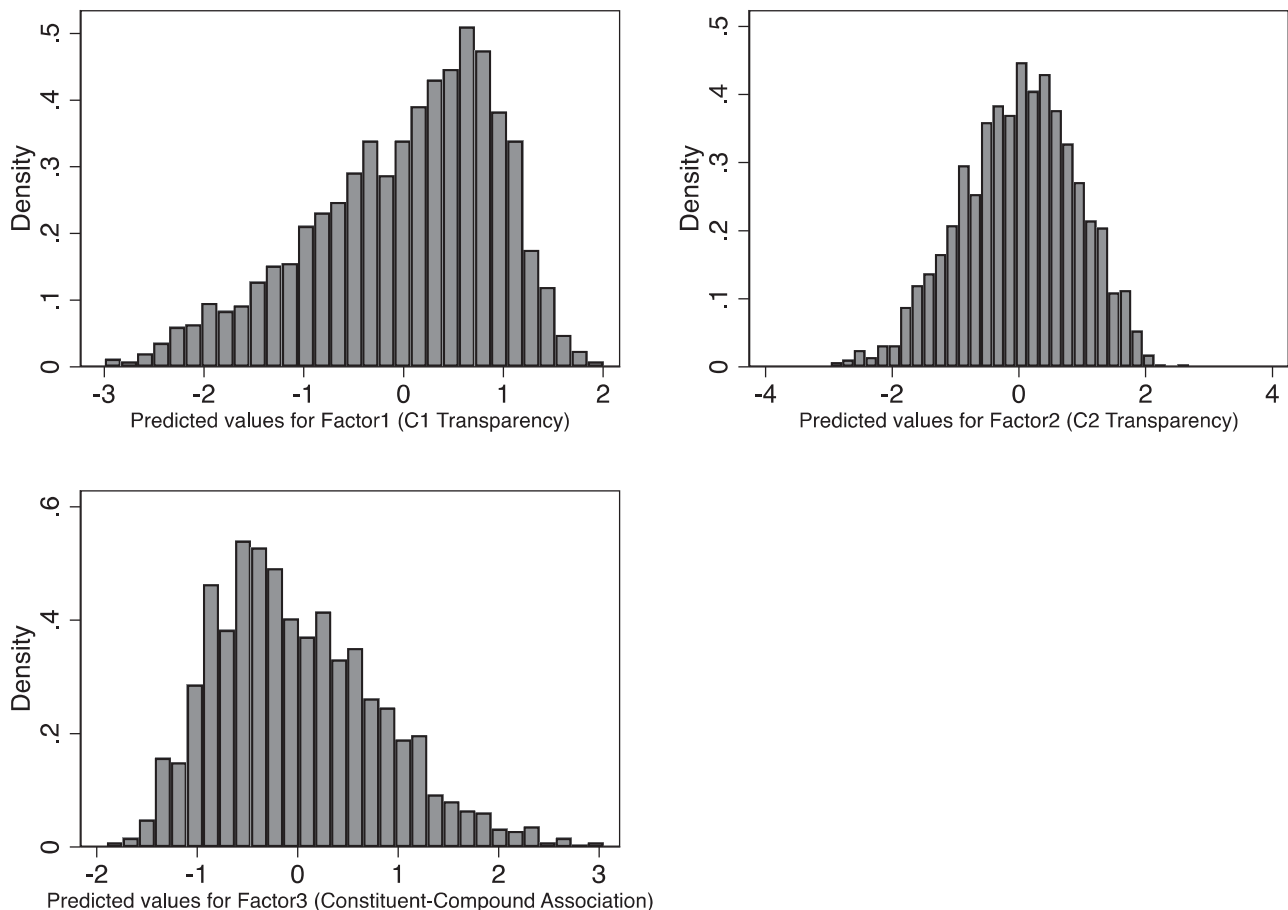


Fig. 4. Histogram of predicted values of Factor 1 (C1 Transparency), Factor 2 (C2 Transparency), and Factor 3 (Constituent-Compound Association) from our factor analysis for the items used in the current project.

Table 7

Standardized regression coefficients with standard errors (in parentheses) from the Analysis 1 models using the nine semantic transparency measures present in LADEC to predict English Lexicon Project lexical decision times and naming latencies, and the British Lexicon Regression project lexical decision times.

	Model 1 elp_ld_rt	Model 2 BLPr	Model 3 elp_naming_mean_rt
ratingcmp	−3.428*** (0.354)	−2.301*** (0.303)	−2.267*** (0.261)
ratingC1	2.498*** (0.267)	1.850*** (0.223)	1.283*** (0.197)
ratingC2	2.396*** (0.271)	1.373*** (0.233)	1.408*** (0.200)
LSAc1stim	−70.67*** (17.860)	−33.16* (15.111)	−37.44** (13.198)
LSAc2stim	−69.01*** (19.489)	−54.31*** (15.919)	−41.81** (14.402)
c1stim_snaucos	−64.55* (27.754)	−5.904 (23.647)	−52.08* (20.509)
c2stim_snaucos	−68.93** (26.213)	−58.62** (22.265)	−47.55* (19.370)
st_c1_mean	−23.32*** (5.254)	−23.20*** (4.565)	−6.426 (3.883)
st_c2_mean	−27.47*** (5.095)	−14.57*** (4.357)	−12.20** (3.765)
_cons	980.9*** (36.202)	809.0*** (31.865)	816.8*** (26.752)
N	1555	1054	1555
Adj R ²	0.126	0.142	0.073
AIC	18806.84	11971.93	17866.03
BIC	18860.33	12021.53	17919.52

*p < 0.05, **p < 0.01, ***p < 0.001.

2019; Kim et al., 2018; Wang et al., 2014), they have yet to be evaluated in a single regression model. Table 7 shows the outcome of the models.

For the model predicting ELP lexical decision times (Table 7, Model 1), all the variables are successful predictors. For the model predicting BLP lexical decision response times (Table 7, Model 2), all variables are successful predictors except the SNAUT association between the first constituent and the compound (c1stim_snaucos). For the model predicting ELP naming latencies (Table 7, Model 3), all are successful predictors except for the Kim et al. (2018) ratings of relatedness between the compound and the first constituent (st_c1_mean). Taken together, the data show that the common variables of semantic transparency are simultaneously useful for predicting both lexical decision times and naming latencies.

While these variables are clearly quite predictive, we recognize that

Table 8

Standardized regression coefficients with standard errors (in parentheses) from the Analysis 1 models using the three human rating variables obtained in LADEC to predict English Lexicon Project lexical decision times and naming latencies, and the British Lexicon Regression project lexical decision times.

	Model 1 elp_ld_rt	Model 2 BLPr	Model 3 elp_naming_mean_rt
ratingcmp	−3.794*** (0.362)	−2.785*** (0.308)	−2.375*** (0.262)
ratingC1	1.877*** (0.240)	1.245*** (0.205)	1.128*** (0.173)
ratingC2	1.619*** (0.235)	1.075*** (0.202)	1.080*** (0.170)
_cons	776.4*** (13.387)	666.8*** (11.286)	695.9*** (9.693)
N	1555	1054	1555
Adj R ²	0.067	0.081	0.050
AIC	18902.66	12039.37	17898.55
BIC	18924.05	12059.21	17919.95

***p < 0.001.

most researchers are unlikely include all nine semantic transparency variables included in this analysis when analyzing experimental data. Instead, they will likely choose a smaller set of variables (e.g., the three LADEC ratings OR the SNAUT values between the constituents and the compound, but not both) to represent semantic transparency. To address this issue, we include a model that can be used for comparison, and uses the rating variables obtained in LADEC (ratingcmp, ratingC1, ratingC2; see Table 1 for details) to predict the ELP and BLP response times. We chose this model as the analyses in Gagné et al. (2019) demonstrate that, of the semantic transparency variables present in LADEC, this was the best-fitting model for predicting the ELP and BLP response times. The results are presented in Table 8. All three of these semantic transparency rating variables are successful predictors of all three ELP and BLP response time variables.

3.2.2. Analysis 2: the three factor values as predictors

The analyses in the preceding section indicated that human-based and corpus-based measures of semantic transparency were successful predictors of lexical decision and naming. In the current section, we examine whether the factors that are composed of these variables are also predictive.

We used the three extracted factors from Model 2 in as predictors for ELP and BLP lexical decision and naming response times. Table 9 shows these results. For the model predicting BLP lexical decision times (Table 9, Model 2), all three factors are successful predictors. For the model predicting ELP lexical decision times (Table 9, Model 1), only the Transparency of the Second Constituent (Factor 2) and the Association between the Constituents and the Compound (Factor 3) are successful predictors. For the model predicting ELP naming latencies (Table 9, Model 3), only the Transparency of the First Constituent (Factor 1) and the Transparency of the Second Constituent (Factor 2) are successful predictors. In other words, the transparency of the second constituent (Factor 2) is predictive of all three response time variables, while the human ratings of the transparency of the first constituent (Factor 1), and the corpus-based computational association between the constituents and compounds (Factor 3) are only predictive of certain response time variables. This suggests that the transparency of the second constituent has a prominent role in determining response time. This idea is corroborated from other studies which show that the transparency of the second constituent is a strong predictor of response time, especially in words where the second constituent designates the semantic and lexical category (Libben et al., 2003; Ji et al., 2011; Marelli and Luzzatti, 2012; Zwitserlood, 1994).

Table 9

Standardized regression coefficients with standard errors (in parentheses) from the Analysis 2 models using the three factors from our factor analysis to predict English Lexicon Project lexical decision times and naming latencies, and the British Lexicon Regression project lexical decision times.

	Model 1 elp_ld_rt	Model 2 BLPr	Model 3 elp_naming_mean_rt
Transparency of the first Constituent (Factor1)	−6.015 (3.191)	−7.862** (2.696)	−5.873* (2.308)
Transparency of the Second Constituent (Factor2)	−8.811** (3.312)	−6.021* (2.768)	−5.103* (2.395)
Association between the Constituents and the Compound (Factor3)	−11.04** (3.922)	−10.70** (3.306)	−0.691 (2.836)
_cons	772.5*** (2.733)	666.8*** (2.291)	695.9*** (1.977)
N	1555	1054	1555
Adj R ²	0.024	0.048	0.009
AIC	18972.19	12076.32	17964.05
BIC	18993.59	12096.16	17985.44

*p < 0.05, **p < 0.01, ***p < 0.001.

Surprisingly, factors comprised of variables which predict lexical decision times and naming latencies, as shown in Analysis 1, are not always predictive. Further, there is no correspondence in factor loadings between the few variables that did not predict response times in Analysis 1, and the factors which did not predict response times in Analysis 2. The SNAUT value between the first constituent and the compound (c1stim_snautCos) loads onto Factor 3 (Association between the Constituents and the Compound) in Model 2, and did not predict BLP response times; however, Factor 3 was predictive of BLP response times. Similarly, human ratings of the relatedness between the first constituent and the compound (st_c1_mean) loads onto Factor 1 (Transparency of the First Constituent) in Model 2, and did not predict ELP naming latencies; but Factor 1 is predictive of ELP naming latencies. This pattern suggests the factors may function differently than individual variables. Also, there may be interactions between the factors which are not accounted for here, but that may serve as better predictors for response time. Analyses 3 and 4 investigate the nature of these potential interactions.

3.2.3. Analysis 3: factor 1 and factor 2 and their interaction as predictor variables

Previous studies have shown that the transparency of the individual constituents within the compound interact in their influence on processing in lexical decision tasks and semantic priming (El-Bialy et al., 2013; Libben et al., 2003; Marelli and Luzzatti, 2012; Schmidtke et al., 2018b; Coolen et al., 1993). In terms of our factors, Factor 1 is representative of the first constituent, whereas Factor 2 is representative of the second constituent. In the current analysis, we investigate whether the factors replicate this first constituent and second constituent interaction by fitting a regression model where the main effects and interaction of the first two factors are predictors of the ELP and BLP variables. We did not include Factor 3 (Association between the Constituents and the Compound) to investigate this question, because it includes significant variable loadings related to both constituents. The results of this analysis are presented in Table 10. The interaction between the Transparency of the First Constituent (Factor 1) and the Transparency of the Second Constituent (Factor 2) is not a successful predictor of any of the three response time variables, but the main effects of each factor are significant in all cases (Table 10, Model 1, Model 2, Model 3). This suggests that both the human ratings related to the first constituent (Factor 1) and the transparency of the second constituent (Factor 2) influence response times for these tasks, but that these constructs do not interact while exerting their influence. Considering previous findings of the interaction between the transparency of the first and second constituent for single variables (El-Bialy et al., 2013; Ji et al., 2011; Libben et al., 2003), the

Table 10

Standardized regression coefficients with standard errors (in parentheses) from the Analysis 3 models using the first two factors from our factor analysis and their interaction to predict English Lexicon Project lexical decision times and naming latencies, and the British Lexicon Regression project lexical decision times.

	Model 1	Model 2	Model 3
	elp_ld_rt	BLPrt	elp_naming_mean_rt
Transparency of the First Constituent (Factor1)	−8.994** (3.009)	−11.35*** (2.514)	−6.079** (2.171)
Transparency of the Second Constituent (Factor2)	−12.750*** (3.102)	−9.197*** (2.590)	−5.306* (2.238)
Factor1 x Factor2	−1.960 (3.189)	2.295 (2.640)	0.150 (2.300)
_cons	772.9*** (2.807)	666.0*** (2.369)	695.9*** (2.025)
N	1555	1054	1555
Adj R ²	0.019	0.039	0.009
AIC	18979.74	12086.02	17964.1
BIC	19001.14	12105.86	17985.5

*p < 0.05, **p < 0.01, ***p < 0.001.

lack of interaction between these factors is surprising; however, it further supports the idea that the factors, which are a representation of the interrelationships between variables, may function differently than the individual variables.

As mentioned, Factor 3, Association between the Constituents and the Compound, was not included in this analysis. Given that these three factors are derived from the same factor model, it is possible that Factor 3 may influence the Factor 1 by Factor 2 interaction. This could explain why the interaction in Analysis 3 is not significant. We consider this potential three-way interaction in Analysis 4.

3.2.4. Analysis 4: allowing for a three-way interaction for factors 1, 2 and 3

Here we examine a regression model predicting the ELP and BLP response time data with all three factors and each possible interaction as predictors. The results are presented in Table 11. There was a significant three-way interaction between the factors in all three regression models, one for each of the three response time variables from the ELP and BLP (Table 11). The p-values for this three-way interaction as a predictor in each respective model were $p = 0.014$ in the model predicting ELP lexical decision latencies (Table 11, Model 1), $p = 0.005$ in the model predicting BLP lexical decision latencies (Table 11, Model 2), and $p = 0.048$ in the model predicting ELP naming latencies (Table 11, Model 3). The nature of this interaction as a predictor is extremely similar for all three response time variables. The margins plot for three-way interactions are shown in Fig. 5 for the model predicting the ELP lexical decision (elp_ld_rt; Table 11, Model 1), Fig. 6 for the model predicting BLP lexical decision (BLPrt; Table 11, Model 2), and Fig. 7 for the model predicting ELP naming latencies (elp_naming_mean_rt; Table 11, Model 3).

The lower panels of Figs. 5–7 show that if the Transparency of the Second Constituent (Factor 2) is sufficiently high, the effect of the Association between the Constituents and the Compound (Factor 3) on response time does not change across the values of the Transparency of the First Constituent (Factor 1). This suggests the other two factors do not observably influence each other if the second constituent is sufficiently transparent (i.e., if Factor 2 is sufficiently high). The upper panels of Figs. 5–7 show that the nature of the interaction changes as the

Table 11

Standardized regression coefficients with standard errors (in parentheses) from the Analysis 4 models using the three factors from our factor analysis and all possible interactions to predict English Lexicon Project lexical decision times and naming latencies, and the British Lexicon Regression project lexical decision times.

	Model 1	Model 2	Model 3
	elp_ld_rt	BLPrt	elp_naming_mean_rt
Transparency of the First Constituent (Factor1)	−13.84*** (3.515)	−14.75*** (2.999)	−10.90*** (2.549)
Transparency of the Second Constituent (Factor2)	−10.38** (3.586)	−7.511* (3.024)	−5.621* (2.600)
Factor1 x Factor2	12.96** (3.948)	13.51*** (3.350)	9.890*** (2.863)
Association between the Constituents and the Compound (Factor3)	−2.953 (4.152)	−4.629 (3.486)	4.61 (3.011)
Factor1 x Factor3	−20.75*** (4.532)	−17.18*** (3.696)	−13.12*** (3.287)
Factor2 x Factor3	−16.32*** (3.948)	−9.707** (3.302)	−10.96*** (2.863)
Factor1 x Factor2 x Factor3	9.449* (3.859)	8.537** (3.065)	5.534* (2.798)
_cons	782.2*** (3.125)	673.5*** (2.639)	701.9*** (2.266)
N	1555	1054	1555
Adj R ²	0.047	0.075	0.028
AIC	18938.46	12049.32	17939.20
BIC	18981.25	12089.00	17981.99

*p < 0.05, **p < 0.01, ***p < 0.001.

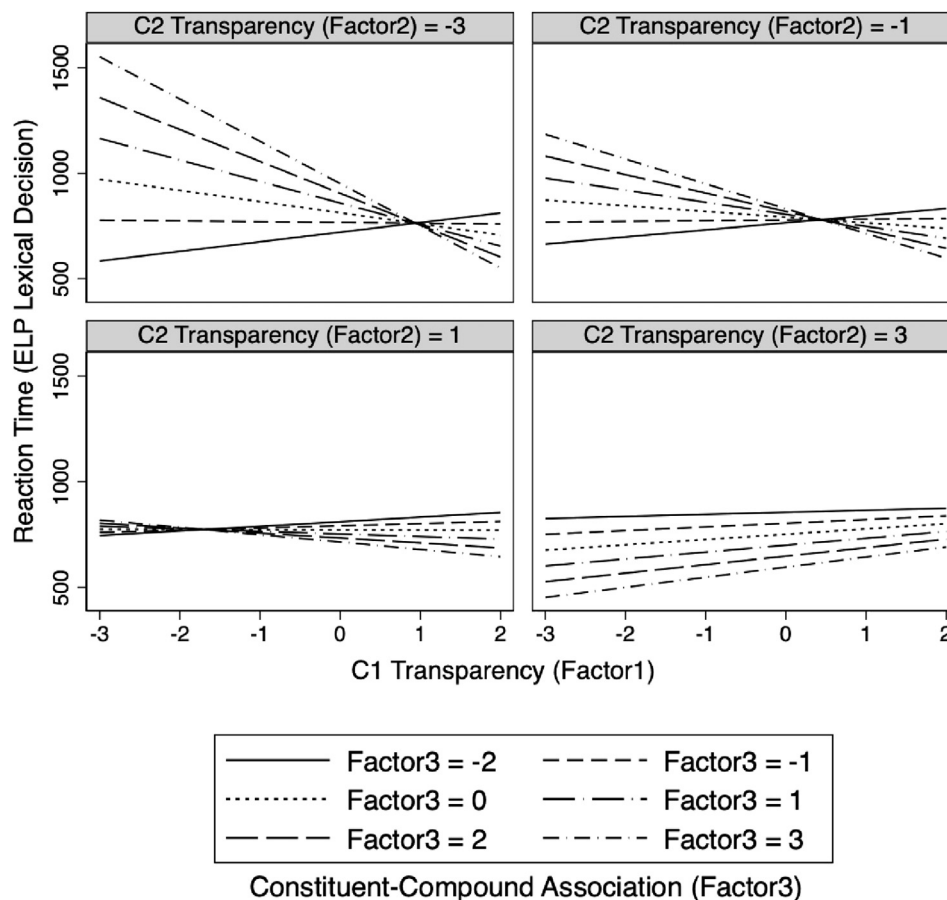


Fig. 5. Graph depicting the three-way interaction among Factor 1 (C1 Transparency) Factor 2 (C2 Transparency) and Factor 3 (Constituent-Compound Association) when predicting the lexical decision response times from the English Lexicon Project (elp_ld_rt) for the items used in the current project. Negative values indicate lower transparency, while positive values indicate higher transparency.

transparency of the second constituent decreases. In this case, the effect of the Association between the Constituents and the Compound (Factor 3) does change across the values of the human ratings related to the Transparency of the First Constituent (Factor 1). If the first constituent is rated as transparent (i.e., if Factor 1 is high), different strengths of association between the constituent and the compound (i.e., any value of Factor 3) will have approximately the same effect on response time. If the first constituent is rated as opaque (i.e., Factor 1 is low), in addition to the second constituent being opaque (i.e., when Factor 2 is low), a stronger association between the constituent and the compound (i.e., higher Factor 3) will be detrimental to response time, while a weaker association will not differentially affect response time.

The interaction shown here seems counterintuitive because one might expect that a stronger association (which corresponds to a more transparent relationship between the constituent and the compound) should be helpful and thus result in faster response times. We propose that this is not the case, because this stronger association disagrees with the other aspects of semantic transparency. When both the Transparency of the First Constituent (Factor 1) and the Transparency of the Second Constituent (Factor 2) are low, the compound would be considered opaque. However, if at the same time there is a strong association between the constituents and the compound, this alternatively suggests that the compound is transparent. Thus, there is a discrepancy between the level of transparency predicted by different constructs. Resolving this discrepancy requires additional processing time and subsequently slows response times.

For example, in *cloudburst*, the composite factors disagree such that the strength of association between the constituents and the compound

predicts a higher level of transparency (i.e., Factor 3 exhibits a positive value) than the level of transparency predicted by the Transparency of the First Constituent and Transparency of the Second Constituent (i.e., both Factor 1 and Factor 2 exhibit negative values). The lexical decision times and naming latencies for *cloudburst* are relatively slow, being 928 ms for the ELP lexical decision, 844 ms for the BLP lexical decision, and 790 ms for the ELP naming. Compare this to the compound *wayside* which is predicted as opaque by a weaker association between the constituents and the compound, lower values for the human ratings of the transparency for the first constituent, and lower values for the transparency of the second constituent (i.e., Factor 1, Factor 2, and Factor 3 all exhibit negative values). The lexical decision times and naming latencies for *wayside* are relatively fast, being 631 ms for the ELP lexical decision, 649 ms for the BLP lexical decision, and 599 ms for the ELP naming. This unified conception of opacity may explain why a weaker association contributes to faster response times, as prompted by *wayside*, and conversely the disjointed conception of opacity may explain why a stronger association contributes to slower response times, as prompted by *cloudburst*.

In addition to showing the practical effects of the factors on response time, this further suggests that there are different aspects of semantic transparency being represented in this analysis. Moreover, discrepancies between the level of transparency anticipated by these aspects are potentially detrimental for processing.

4. General Discussion

Within the psycholinguistic literature on the processing and representation of morphologically complex words, semantic transparency has

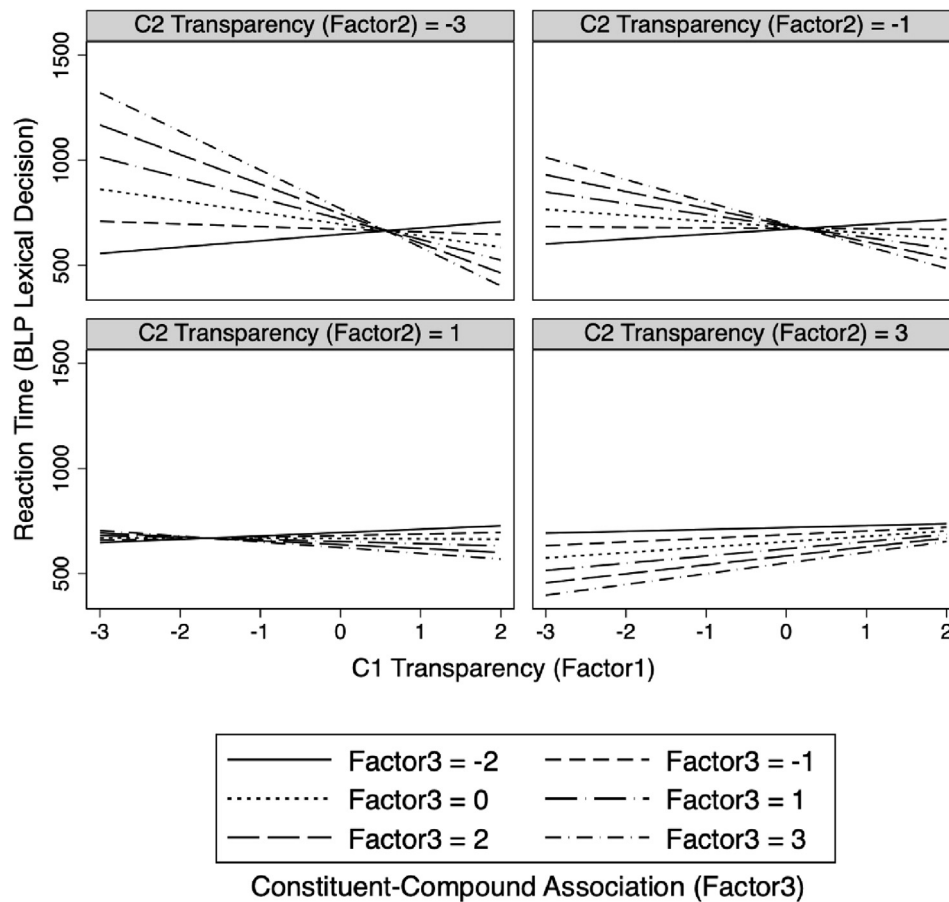


Fig. 6. Graph depicting the three-way interaction among Factor 1 (C1 Transparency) Factor 2 (C2 Transparency) and Factor 3 (Constituent-Compound Association) when predicting the lexical decision response times from the British Lexicon Project (BLPrt) for the items used in the current project. Negative values indicate lower transparency, while positive values indicate higher transparency.

often been construed as being a single construct, although some researchers have suggested that semantic transparency might have several aspects (Gagné et al., 2019; Marelli and Luzzatti, 2012). Our results shed light on this theoretical issue in that they indicate that semantic transparency has multiple aspects; the common operational measures of semantic transparency do not map onto a single, unified construct; the eleven operational measures of semantic transparency originally analyzed in this study informed four factors, each representing a different aspect or notion of semantic transparency. We found that the predicted values for the first three factors related to the human ratings of the transparency of the first constituent (Factor 1), the human ratings of the transparency of the second constituent and its association to the compound (Factor 2), and the association between the constituents and the compound (Factor 3). Furthermore, we found that these factors display a three-way interaction when predicting response times for the ELP and BLP lexical decision and naming data. Factor 4, representing the association between the constituents (excluding the compound), seemed to be targeting a construct which the other factors were not, and so was removed from subsequent analyses.

4.1. Distinction between human ratings and corpus-based association measures

Semantic transparency has been operationalized in terms of either corpus-based or human ratings. Our findings suggest that these two methods of measuring semantic transparency are not interchangeable and, thus, the conclusions drawn from existing work about whether

semantic transparency influences processing should be interpreted in light of which type of measure has been used. Our analyses demonstrate that human ratings of relatedness, retention, and predictability do not inform the same aspect of transparency as corpus-based measures of association. The factor representing the first constituent transparency ratings (Factor 1) and the factor representing the association between the constituents and the compound (Factor 3) clearly show the differentiation between these two types of measures for the first constituent. This corroborates the differential influences on processing between corpus-based and human measures previously found, which suggested separate facets of semantic transparency are represented by these measures (Gagné et al., 2016; Schmidtke et al., 2018b; Marelli and Luzzatti, 2012). Further, it suggests that human participants are sensitive to aspects of compound transparency, especially those related to the first constituent, that corpus-based computational measures, such as LSA and SNAUT, are not. Because of this, researchers interested in the study of semantic transparency may need to account for this distinction when comparing studies that utilize different types of measures, and when choosing different measures for their own research.

In our factor analysis, a fourth factor was present in Model 1 which informed a separate construct from the other three factors. This fourth factor was comprised solely of two variables which represented the association between the constituents, absent from the compound. We propose that these variables loaded separately because they inherently do not account for relationships with the entire compound. That is, whether *shin* or *dig* are similar may not be directly relevant to the meaning and processing of *shindig*. This factor representing the

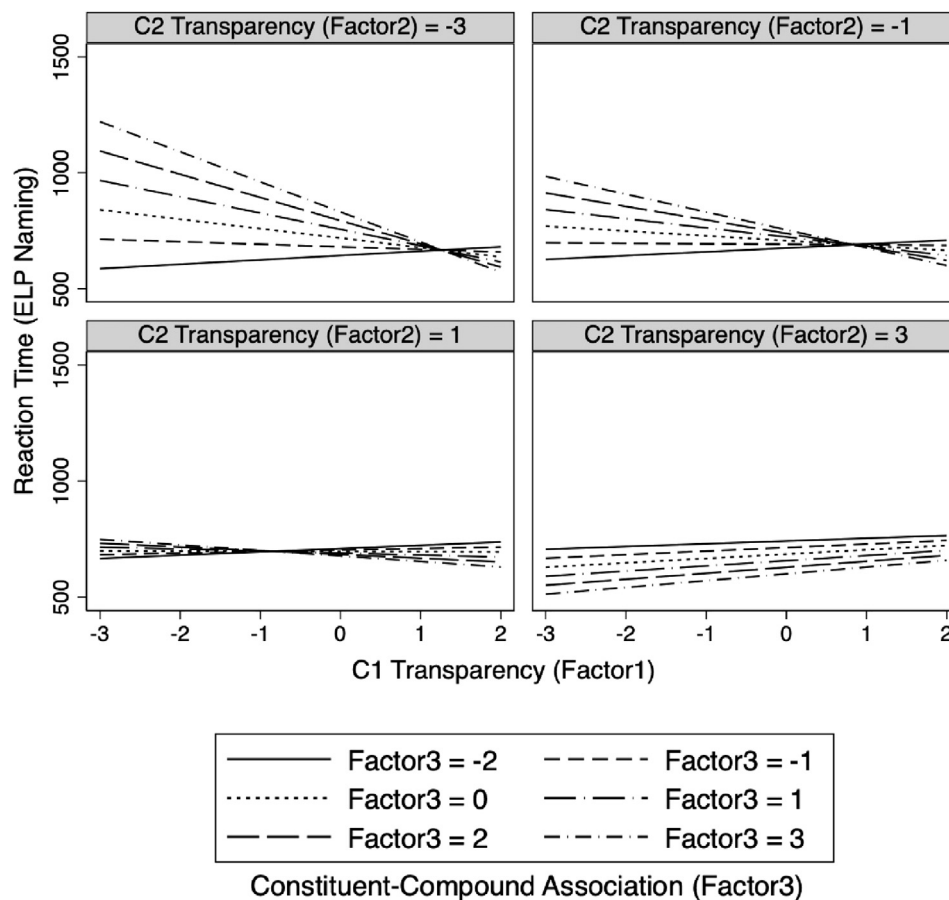


Fig. 7. Graph depicting the three-way interaction among Factor 1 (C1 Transparency) Factor 2 (C2 Transparency) and Factor 3 (Constituent-Compound Association) when predicting the naming latencies from the English Lexicon Project (elp_naming_mean_rt) for the items used in the current project. Negative values indicate lower transparency, while positive values indicate higher transparency.

association between the constituents alone illustrates that, while corpus-based measures differ from human ratings, corpus-based measures may also differ from each other. Corpus-based computational measures that include the compound itself may be a closer representation of semantic transparency than those that do not.

4.2. First and second constituent contributions

The current findings address the theoretical question of whether it is necessary to have separate measures of transparency for each constituent or whether it is sufficient to have only one measure for the entire compound (e.g., a rating that asks participants to indicate whether the two constituents were “transparently related to the meaning of the compound” as in Juhász et al., 2015). Our factor analyses indicate that each constituent uniquely influences semantic transparency and its influence on processing. We observed that the first factor relates to the transparency of the first constituent, while the second factor relates to the transparency of the second constituent. The composition of each of these constituent factors is also unique: the Transparency of the Second Constituent (Factor 2) is comprised of both corpus-based and human rating variables related to the second constituent, while the Transparency of the First Constituent (Factor 1) is only composed of human rating variables related to the first constituent. The corpus-based computational measures related to the first constituent and the compound comprise the third Factor, and the corpus-based computational measures related to the second constituent and the compound secondarily load onto this factor as well (these variables primarily load onto the Transparency of the Second Constituent). Thus, this third Factor is a computational Factor that accounts for the corpus-based association between both constituents and

the compound, but primarily between the first constituent and the compound. The differences between the constituents was also supported by Factor Model 3 (section 2.2.3), where we forced a two-factor solution. In this model, the variables organized themselves based on whether they were related to the first or the second constituent, rather than the type of measure used. This shows the distinction between the constituents is a more powerful differentiator between variables than is the specific type of operational measure. Clearly, each constituent has a unique influence on the level of transparency of the compound.

Our results also suggest the first and second constituents differentially affect the processing of English compounds, and that the role played by the constituents matters. For example, in Regression Analysis 2 (section 3.2.2), where we looked solely at the main effects of the factors without allowing for interactions, the transparency of the second constituent was predictive of all three response time variables while the human rating of the transparency of the first constituent and the association between the constituent and the compound were only predictive of some of the response time variables. These results suggest that there is a distinction between the effects of the first constituent's transparency and the second constituent's transparency on processing English compounds, and that the transparency of the second constituent may be especially predictive of processing effort for visually represented lexical decision and naming tasks.

That each constituent differentially affects the processing of English compounds is supported by previous research which shows that the constituents behave differently when predicting human behaviour. For example, Libben et al. (2003) analyzed semantic transparency in English compounds and found that compounds with opaque second constituents were more difficult to process than compounds with transparent second

constituents, regardless of the transparency of the first constituent. That is, the transparency of the first constituent did not have the same influence on compound processing as the transparency of the second constituent. Our results show a similar finding, as the transparency of the second constituent was consistently predictive of common response time variables from the ELP and BLP. Other research has also found differences between effects of the first and second constituents' transparency (e.g., Marelli and Luzzatti, 2012; Schmidtke et al., 2018b).

In addition to differentially affecting semantic transparency and its effect on processing, the constituents also differ in their role in the compound. In English, the second constituent often designates both the semantic and lexical category of the compound (i.e., the second constituent is the head of the compound; Lieber, 1981; Scalise and Fabregas, 2010; Williams, 1981). This is particularly true for transparent compounds, or partially opaque compounds with transparent second constituents (Libben et al., 2003). For example, *blueberry* (a Transparent-Transparent compound) and *strawberry* (an Opaque-Transparent compound) are a type of berry, while *shindig* is not a 'type of dig' or method of digging. This "type-of" relation is known as hyponymy (Lyons, 1977; Chaffin and Glass, 1990; Marelli et al., 2015; Reeves et al., 1993), and has been shown to be intimately related to semantic transparency as it relates to the compound and each of its constituents (Gagné et al., 2020).

Given that the two constituents differ in their effects on compound processing, why do the corpus-based measures of the second constituent load with the human ratings of transparency for the second constituent, but the corpus-based measures of the first constituent not load with the human ratings of transparency for the first constituent? First, it is important to recall why the corpus-based variables, which reflect the contexts in which two words occur, serve as a proxy for semantic transparency and semantic similarity. They have been used as a proxy for semantic similarity because words that are semantically similar, or come from similar semantic categories, are likely to be used in similar contexts. Because the head generally designates the semantic category of the compound and English has only head-final compounds (other than a handful of borrowed items such as *fettucine alfredo*), this measure of similarity indicates that a transparent second constituent is used in a context more like that of the compound than is an opaque second constituent. For example, the contexts in which *blueberry* is used is likely more similar to the contexts where *berry* is used, while the contexts for *shindig* is likely less similar than the contexts for *dig*. As a result, the association between a transparent second constituent and the compound would be higher for corpus-based computational measures such as LSA or SNAUT (Kuperman, 2013; Wang et al., 2014; Landauer and Dumais, 1997; Mandera et al., 2017). The corpus-based computational measures related to the second constituent load onto Factor 2 because this constituent is used in a context that is similar to the overall compound (Wang et al., 2014; Marelli and Luzzatti, 2012), precisely because the transparent second constituent largely corresponds to a clear semantic head-ness relationship, and this relationship drives both the human judgment of transparency and its representation in distributional semantics. One possible reason for this finding is that the human judgments of the semantic transparency of the second constituent largely reflects the degree of hyponymy (Gagné et al., 2020). Semantic transparency of the first constituent, on the other hand, is not as strongly related to hyponymy. This has implications for the ability of these measures to represent each constituent. While a corpus-based computational measure may be sufficient to represent the effect of the transparency of the second constituent on processing in English (due to its being a head-final language), this same type of measure may not be adequate for representing the effect of the transparency of the first constituent, or modifier, on processing.

4.3. Factors in relation to human performance

Our goal was to examine the relationships among the various measures of semantic transparency and to understand how these relations

inform our understanding and representation of semantic transparency as a theoretical construct. Individual variables of semantic transparency are targeted (and hence, partial) representations of this construct and sets of these variables have been demonstrated to be useful in evaluating human performance both here and elsewhere (Gagné et al., 2019; Kim et al., 2018; Juhasz et al., 2015). Still, although undoubtedly informative because these individual variables are partial representations of semantic transparency, the failure of an individual variable as, for example, a predictor of lexical decision times does not necessarily mean that semantic transparency does not affect lexical decision. In short, because the variables differ in how they instantiate semantic transparency, different variables are likely to be differentially related to human performance. Factors are generally used to more concretely define abstract constructs, such as semantic transparency, by representing the interrelationships between multiple variables related to the construct.

As we saw, the Factors are themselves useful for predicting response time in lexical decision and naming, similar to individual variables or variable sets. After regressing both the factors and the set of variables which compose them with common response time variables from the ELP and BLP, it appears that composite factors are roughly comparable to the set of individual variables when predicting response times. The benefit in using factors, rather than a large set of variables, is related to simplicity. A regression model with three predictors (e.g., Analysis 2) is far more parsimonious than a regression model with nine predictors (e.g., Analysis 1, Table 7). As a result, the composite Factors may be particularly useful when comparing the effect of singular, broader aspects, such as the transparency of the second constituent. These aspects may require the inclusion of multiple single predictor variables in a regression analysis or statistical model, but they could be captured by a single Factor and subsequently analyzed.

However, perhaps the most pressing question researchers might be wondering about at this point is "Which variables, then, should be used to operationalize semantic transparency?" As an attempt to answer this question, we compare the Akaike information criterion (AIC) and Bayesian information criterion (BIC) values from our regression analyses, specifically from Analysis 1 (section 3.2.1), Analysis 2 (section 3.2.2), and Analysis 4 (section 3.2.4). The regression models with the lowest AIC and BIC values, and therefore the best fitting models, were those that had all nine individual semantic transparency variables as predictors (see Table 7 for details). This conclusion was also supported by the adjusted R^2 values. It would be expected that a model with more variables could account for more variability in the data (note, however, that AIC, BIC, and adjusted R^2 do penalize overly complex models), but the inclusion of nine semantic transparency variables is inefficient and unlikely to be practiced by most researchers. Therefore, we look to the regression models which contain fewer variables: the models with the LADEC rating variables (see Table 8 for details), the models with the Factors (see Table 9 for details), and the models with the Factors and their interactions (see Table 11 for details). Of these three models, the least predictive models (i.e., the ones with the highest AIC and BIC values) were those where the only predictors were the individual Factors. The models including the interaction between all three Factors as a predictor were better fitting models, as indicated by AIC and BIC values, than the models lacking this interaction. This is corroborated by the improved adjusted R^2 values in the models including the three-way Factor interaction (Table 11) compared to the models without any Factor interactions (Table 9) or with only a two-way interaction between the first two Factors (Table 10). These results emphasize the importance of considering the interactions between the Factors, which represent different aspects of semantic transparency. The models where the LADEC ratings are predictors (Table 8) were better fitting models than were the regression models containing Factor variables according to both AIC and BIC (and similarly corroborated by adjusted R^2). Hence, the human rating variables obtained in LADEC (see Gagné et al., 2019 for details) appear to be particularly useful in predicting response time, being roughly comparable to either the composite factors we obtained or the nine individual semantic transparency variables.

In this study, the effects of semantic transparency are shown in lexical decision and naming tasks through both composite factors and individual variables. Future studies may consider other behavioural tasks which are similarly sensitive to semantic transparency effects. For example, experiments using eye-tracking or other experimental tasks may be further advanced by implementing an evaluation of semantic transparency effects on behavioural data, similar to the one used in the current study. This could be a fruitful approach to investigating semantic transparency in the future because eye-tracking already has been used to examine the potential influence of semantic transparency (e.g., Frisson et al., 2008; Marelli and Luzzatti, 2012; Brusnighan and Folk, 2012; Juhasz, 2007; Schmidtke et al., 2018), and a recent database called CompLex (Schmidtke et al., 2020), which is similar in format to the LADEC database used in the current study (Gagné et al., 2019), contains eye-tracking data from reading compounds in sentence contexts.

5. Conclusions

This study has shown that different operational measures reflect different aspects of semantic transparency of compound words depending on whether they are human ratings or computed from corpus data, and whether they are related to the first or second constituent. The exploratory factor analysis presented in the current study provides a potential structure of the construct “semantic transparency” that might be used for future confirmatory analyses as well as for guiding decisions about which specific measures or perhaps composite factors (i.e., a representation of the interrelationships between a set of measures) are likely to be most useful for particular research questions.

Another underexplored issue in the literature relates to potential interactions among the various measures, and the representative capability of both composite factors and individual variables. We found that the effect of second constituent transparency, human ratings of the first constituent transparency, and the association between the constituent and the compound all interact in the prediction of BLP and ELP response times. This shows the aspects informed by common measures of semantic transparency are indeed different, and behave differently when predicting human processing - but, that they also depend on one another in an interaction. Because different studies measure this construct in different ways, not finding an effect of semantic transparency on processing may be informative in relation to one aspect of semantic transparency (e.g., meaning relatedness) but not another (e.g., constituent meaning retention). That is, semantic transparency could influence processing, but certain individual operational measures may not be as sensitive to displaying these effects. Our results from the regression analyses show that the factors behave differently than individual variables when predicting common response time variables from the ELP and BLP. These results could be extended with the inclusion of other behavioural data, including those which are more naturalistic such as reading in eye-tracking experiments. The results from the current project are especially informative in understanding how the measures represent the underlying aspects of the construct “semantic transparency”, how these potential differences may account for the conflicting results surrounding semantic transparency in the literature, and suggest that using factors constructed from the interrelationships between measures rather than the individual measures themselves might be a fruitful approach in certain situations. Future studies which consider these underlying aspects may be able to unify the currently mixed understandings of how semantic transparency influences processing within the human language system.

6. Availability and open practices statement

The LADEC_SemanticTransparency-FactorAnalysis dataset is available for public download at <https://era.library.ualberta.ca/> (search term:

LADEC_SemanticTransparency-FactorAnalysis), formatted as text (.csv) and as a Stata datafile. When using variables that were obtained from other sources, please cite the original source. This research was not preregistered.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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